# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years of experience in **production troubleshooting**, **Java/C++/Python applications**, root-cause analysis, automated testing, and **reliable customer-facing systems** across cloud and cross-platform environments.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

## Experience

**USC Dornsife** | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production monitoring supporting dependable internal software workflows and technical operations.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, diagnosing performance bottlenecks across normalized schemas and tenant-isolated workflows supporting 500+ production users reliably.

**Grad8** | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production user platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, secure ingestion, AWS RDS, and CI/CD workflows.
- Processed 1K+ files per batch through **multithreaded validation and fault isolation**, reproducing heterogeneous input failures while preserving responsive workflows during concurrent processing and sustained production traffic reliably.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* | *India*
- Improved API response times by 30% through **root-cause debugging**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern - (VibrantChem)* | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform**, designing reusable interfaces for customers, billing, allocation, permissions, and notifications with real-time updates across distributed locations reliably.
- Increased lead conversions by 15% through **customer and lead-management experiences**, integrating role-based access, WebSocket updates, notifications, Azure SQL, and modular services across eight operational product domains reliably.

**LST Capitals** | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.
- Delivered **sub-200ms interactive workflows** across 50K+ booking records, integrating pricing, coupons, payment webhooks, Firebase synchronization, and status updates while preventing conflicting reservations across concurrent user sessions reliably.

## Skills

**Languages:** Java, C++, Python, JavaScript, TypeScript, SQL, Bash
**Software Engineering:** Object-Oriented Programming, Object-Oriented Design, System Design, Data Structures, Algorithms, Software Development Lifecycle, Agile, Scrum
**Backend & Data:** Flask, FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL, Caching, JSON
**Support & Diagnostics:** Root-Cause Analysis, Log Analysis, Splunk, JIRA, Debugging, Issue Tracking, Error Handling, Fault Isolation, Performance Optimization, Monitoring, Technical Documentation
**Testing & Delivery:** Unit Testing, Integration Testing, Automated Testing, Regression Testing, PyTest, Mockito, Jest, Selenium, Jenkins, CI/CD, Git, GitHub, Maven
**Platforms & Protocols:** Azure, AWS, Docker, Linux, Windows, Cross-Platform Development, TCP/IP, WebSockets, Sentry, Postman

## Projects

**HMIS - Hospital Management Information System** | Live
- Built a **healthcare management platform** with TypeScript, ABDM APIs, consent-based patient access, appointment workflows, Twilio notifications, and Sentry monitoring, reducing administrative workload by 30% across operational workflows.

**Chat2SQL** | Live
- Extended a **cross-platform Java application** supporting 10+ database systems, using Docker, Maven, Node.js, and Yarn for structured validation, schema exploration, query workflows, local debugging, and repeatable deployment.

**CPU-Scheduling Simulator** | Live
- Built a **CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios reliably.